COMPUTER NETWORKS ASSIGNMENT

Design and Performance Evaluation of a Next-Gen Smart Campus Network Architecture

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Course Code: CSA0712

Course: COMPUTER NETWORKS

Faculty: Dr.S.Salini

Assignment Submission

1. Understanding of the Given Assignment

The supplied assignment sheet is for CSA0712 – Computer Networks. Its title is “Design and Performance Evaluation of a Next-Gen Smart Campus Network Architecture.” The task requires a scalable multi-tier Smart Campus network, simulation with network tools, comparison of two routing/design alternatives, and evaluation using throughput, packet delivery ratio and end-to-end delay. It also requires topology, IP addressing/subnetting, routing configuration, screenshots/results, QoS and congestion analysis, failure-mode analysis and GitHub documentation.

This document is intentionally prepared as an assignment solution, not as a capstone-project report. The practical implementation is designed for Cisco Packet Tracer and the screenshot checklist identifies the actual screenshots that should be captured from the student's Packet Tracer file.

2. Problem Solution

The proposed solution divides the campus into logical VLANs and IP subnets. Academic users, laboratories, library users, VoIP, CCTV/security, IoT, servers and network management are separated. A multilayer/core layer performs Layer-3 forwarding, while two core routers provide a more reliable routed backbone. OSPF is selected for the improved alternative because it can dynamically learn routes and adapt after a path change. QoS is used to protect delay-sensitive voice traffic from ordinary bulk data.

Main objectives:

- Build the Smart Campus topology in Cisco Packet Tracer.
- Create separate subnets/VLANs and assign correct gateways.
- Demonstrate communication between different subnets.
- Implement and compare two routing/design alternatives.
- Measure throughput, packet delivery ratio, delay and packet loss using repeatable tests.
- Apply QoS concepts for congestion management and real-time traffic.
- Test link/device failures and explain reliability and scalability.

3. Network Architecture and Topology

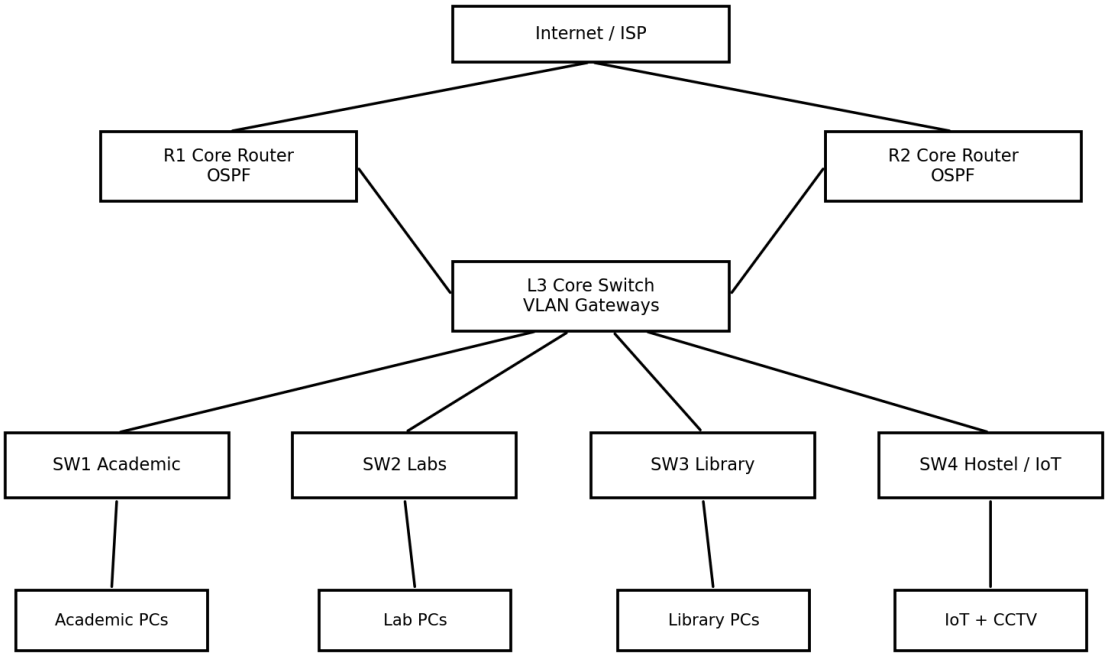


Figure: Smart Campus topology to reproduce in Cisco Packet Tracer

Figure 1. Logical topology for the Cisco Packet Tracer implementation.

Layer/Device	Packet Tracer Device	Role
Internet/Edge	Cloud or generic router	External network representation
Core	2 × Cisco 2911 routers	Inter-network routing and OSPF
Core/Distribution	1 × Cisco 3560 multilayer switch	VLAN gateways and aggregation
Access	4 × Cisco 2960 switches	Connect campus users/devices
End devices	PCs, server, IP phone/IoT/CCTV	Generate application traffic

Communication is primarily unicast for host-to-host traffic. Broadcasts remain within VLANs, reducing unnecessary campus-wide traffic. Ethernet is used for the wired links. The hierarchical design makes the network easier to manage and expand.

4. IP Addressing and Subnetting Plan

VLAN	Purpose	Network	Gateway	Example Host
10	Academic	10.10.10.0/24	10.10.10.1	10.10.10.10
20	Computer Labs	10.10.20.0/24	10.10.20.1	10.10.20.10
30	Library	10.10.30.0/24	10.10.30.1	10.10.30.10
40	VoIP	10.10.40.0/24	10.10.40.1	10.10.40.10
50	CCTV/Security	10.10.50.0/24	10.10.50.1	10.10.50.10
60	IoT	10.10.60.0/24	10.10.60.1	10.10.60.10
70	Servers	10.10.70.0/24	10.10.70.1	10.10.70.10
99	Management	10.10.99.0/24	10.10.99.1	10.10.99.10

A /24 is used in the simulation for simple configuration and future growth. In a production campus, VLSM could allocate smaller prefixes to small groups and larger prefixes to high-density areas.

5. Alternative 1 — Centralized Router-on-a-Stick

In Alternative 1, one router performs inter-VLAN routing through 802.1Q subinterfaces. Access switches connect users through VLANs. This is inexpensive and easy to configure, but the central router and its uplink can become bottlenecks or single points of failure.

```
enable
configure terminal
interface g0/0
 no shutdown
interface g0/0.10
 encapsulation dot1Q 10
 ip address 10.10.10.1 255.255.255.0
interface g0/0.20
 encapsulation dot1Q 20
 ip address 10.10.20.1 255.255.255.0
interface g0/0.30
 encapsulation dot1Q 30
 ip address 10.10.30.1 255.255.255.0
interface g0/0.40
 encapsulation dot1Q 40
 ip address 10.10.40.1 255.255.255.0
end
write memory
```

6. Alternative 2 — OSPF Multi-Tier Design

Alternative 2 introduces two core routers and dynamic OSPF routing. OSPF advertises reachable networks and can recalculate routes after a topology change. The design reduces dependence on one routing device and is more appropriate for a growing smart campus.

```
! R1
enable
configure terminal
router ospf 10
 router-id 1.1.1.1
 network 10.10.10.0 0.0.0.255 area 0
 network 10.10.20.0 0.0.0.255 area 0
 network 10.10.30.0 0.0.0.255 area 0
 network 10.10.40.0 0.0.0.255 area 0
end

! R2
enable
configure terminal
router ospf 10
 router-id 2.2.2.2
 network 10.10.50.0 0.0.0.255 area 0
 network 10.10.60.0 0.0.0.255 area 0
 network 10.10.70.0 0.0.0.255 area 0
 network 10.10.99.0 0.0.0.255 area 0
end
write memory
```

Important: the exact interface IPs and OSPF network statements must match the topology actually built in Packet Tracer. After configuration, verify using show ip interface brief, show ip route and show ip ospf neighbor.

7. Alternative Comparison

Criterion	Alternative 1	Alternative 2

Routing	Static/router-on-a-stick	Dynamic OSPF
Complexity	Low	Moderate
Scalability	Moderate	High
Fault tolerance	Low	Higher
Congestion risk	Higher at central point	Lower with hierarchical design
Route adaptation	Manual/static changes	Dynamic reconvergence
Best use	Small/simple network	Smart campus with growth
Recommendation	Baseline	Preferred

8. Cisco Packet Tracer Configuration Steps

1. Place the routers, multilayer switch, access switches and end devices shown in Figure 1.
2. Connect devices using appropriate copper/optical links available in Packet Tracer.
3. Create VLANs 10, 20, 30, 40, 50, 60, 70 and 99.
4. Assign end-device switch ports to the correct VLANs.
5. Configure inter-switch links as trunks where VLANs must cross the link.
6. Assign gateway addresses to the Layer-3 interfaces/subinterfaces.
7. Configure Alternative 1 and test connectivity.
8. Save a separate Packet Tracer file for Alternative 2 and configure OSPF.
9. Verify OSPF neighbours and routes.
10. Use ping/traceroute and Simulation Mode to demonstrate packet movement.
11. Apply the QoS policy supported by the selected Packet Tracer IOS image.
12. Repeat the same traffic tests for both alternatives and record measured results.

9. Verification Commands

```
show ip interface brief
show vlan brief
show interfaces trunk
show ip route
show ip ospf neighbor
show ip ospf interface
show policy-map interface
ping 10.10.20.10
ping 10.10.70.10
traceroute 10.10.70.10
```

10. QoS and Congestion Management

QoS separates important real-time traffic from ordinary data. VoIP is treated as delay-sensitive, CCTV as sustained high-bandwidth traffic, and normal academic/web traffic as best-effort. A Cisco MQC policy can classify traffic and give priority service to real-time packets. Packet Tracer IOS feature support varies, so the commands must be checked on the selected router.

```
class-map match-any REALTIME
 match access-group 110
!
access-list 110 permit udp any any range 16384 32767
!
policy-map CAMPUS-QOS
 class REALTIME
  priority percent 20
 class class-default
  fair-queue
!
interface g0/0
 service-policy output CAMPUS-QOS
```

The purpose of the policy is to reduce queueing delay for real-time traffic when congestion occurs. Verification should be performed with show policy-map interface.

11. Performance Evaluation

Both alternatives must be tested under the same conditions. Use the same source/destination pairs, device count and traffic pattern. The final numerical values below must be replaced with the actual values observed in the student's Cisco Packet Tracer simulation; this prevents fabricated experimental data.

Metric	How to evaluate	Expected interpretation
Throughput	Delivered data over a fixed interval	Higher is better for bulk traffic
Packet Delivery Ratio	Received packets / sent packets × 100	Closer to 100% is better
End-to-End Delay	Source-to-destination packet time	Lower is better for VoIP/interactive traffic
Packet Loss	Sent minus received packets	Lower is better
Congestion	Queue/load behaviour under high traffic	Less persistent queueing is better
QoS effectiveness	Compare real-time traffic before/after policy	Priority traffic should improve

12. Results Table — Enter Packet Tracer Measurements

Metric	Alternative 1	Alternative 2	Better
Throughput	_____	_____	_____
Packet Delivery Ratio	_____	_____	_____
Average Delay			
Packet Loss	_____	_____	_____
Congestion observation	_____	_____	_____
QoS observation	_____	_____	_____

13. Required Cisco Packet Tracer Screenshots

- Screenshot 1: Complete topology with all routers, switches and end devices.
- Screenshot 2: VLAN configuration (show vlan brief) and trunk configuration.
- Screenshot 3: IP interface status (show ip interface brief).
- Screenshot 4: Routing table (show ip route).
- Screenshot 5: OSPF neighbour relationship (show ip ospf neighbor).
- Screenshot 6: Successful inter-subnet ping.
- Screenshot 7: Simulation Mode showing packet movement from one subnet to another.
- Screenshot 8: QoS configuration and show policy-map interface output.
- Screenshot 9: Performance/congestion test evidence.
- Screenshot 10: Failure test with a routed link disconnected and the resulting path/routing change.

These are screenshot instructions, not fabricated screenshots. Capture them directly from your own Cisco Packet Tracer window after the topology and configurations are working.

14. Failure Mode, Reliability and Scalability Analysis

Failure/Condition	Effect in Alternative 1	Effect in Alternative 2	Mitigation
Central router failure	Major outage	Other core path may remain available	Redundant core
Core link failure	Connectivity may be lost	OSPF can reconverge if an alternate path exists	Redundant routed links
Access switch failure	Local group affected	Local group affected; core remains operational	Spare switch/redundant access
Peak traffic	Central bottleneck likely	QoS and distributed routing reduce impact	QoS, capacity planning
New building	Manual expansion is harder	New VLAN/subnet can be advertised through OSPF	Hierarchical scaling
QoS policy error	Priority service may fail	Same risk, but can be isolated to policy interface	Test and verify policy

The reliability advantage of Alternative 2 depends on actually providing physically/logically redundant paths. OSPF alone cannot recover from a failure if no alternate path exists. Scalability is improved by VLAN segmentation, hierarchical access/distribution/core design and dynamic routing.

15. Final Discussion

Alternative 1 is suitable as a baseline because it is straightforward and has low configuration overhead. However, its centralized routing arrangement is less suitable for a smart campus with many concurrent users and mixed traffic classes. Alternative 2 provides a stronger engineering solution because it separates campus traffic into subnets, supports dynamic routing and provides a basis for redundant paths and QoS.

The proposed design directly addresses the assignment conditions: network topology and device selection, IP addressing and subnetting, inter-subnet routing, comparison of two alternatives, performance evaluation, QoS and congestion management, and failure/reliability/scalability analysis. The empirical performance conclusion should be finalized only after the Packet Tracer tests are executed and the measured values are entered in the results table.

16. Conclusion

A multi-tier Smart Campus network is a practical solution for supporting academic, laboratory, library, VoIP, CCTV, server and IoT services. VLAN-based segmentation controls broadcast domains and improves organization, while OSPF provides dynamic route management. QoS can protect delay-sensitive traffic during congestion. Based on the design characteristics, Alternative 2 is recommended for the final implementation, provided that the required redundant links and QoS policies are configured and verified in Cisco Packet Tracer.

17. GitHub Submission Checklist

- Upload the final .pkt Packet Tracer file.
- Upload this assignment document or PDF if required by faculty.
- Upload topology image and Packet Tracer screenshots.
- Upload a README explaining how to open and test the .pkt file.
- Upload configuration notes/commands and measured results.
- Add the final GitHub repository link in the submission portal.

GitHub Repository: _____

18. Submission Details

Field	Details
Student Name	MUKESH M
Register Number	192521368
Course Code	CSA0712
Course	COMPUTER NETWORKS
Faculty	Dr.S.Salini
Assignment	Design and Performance Evaluation of a Next-Gen Smart Campus Network Architecture
Simulation Tool	Cisco Packet Tracer